

# Predictive Multi-Criteria Allocation across DeFi Lending Protocols: A Forecast-Driven Extension of the AI-Managed ERC-4626 Vault on the `fractal-defi` Substrate

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## Abstract

We study dynamic capital allocation between Aave V3 and Compound V3 USDC supply markets on Ethereum mainnet and ask whether replacing a reactive EMA-smoothed rate observation with a 12-hour-ahead learned forecast yields a significant improvement in risk-adjusted return. The proposed allocator combines a dual-branch DA-BiGRU-CNN forecaster — decomposing each protocol’s supply rate into a deterministic kink component and a learned residual — with a four-factor TOPSIS-style multi-criteria decision rule inherited from the author’s prior ERC-4626 yield vault [19], implemented as a concrete subclass of a new `BaseLendingAllocationStrategy` abstraction contributed to the `fractal-defi` research substrate [12, 13]. We assembled an 18-month hourly panel (13,096 Aave V3 and 12,904 Compound V3 bars, Nov 2024 – Apr 2026); on the 12,895-hour overlap we empirically find a cross-protocol spread with median  $-0.232$  pp, standard deviation  $2.513$  pp, and an Aave-pays-more share of  $40.7\%$ , and we observe a regime shift between 2025 Q3 and 2025 Q4 where this share jumps  $39.9\% \rightarrow 56.3\%$  — the first quarter on the panel where Aave systematically pays more. An initial sparse-sample analysis (37% Compound coverage) had suggested a 2025 Q1→Q2 shift; the 98.5% full-coverage panel revealed that finding to be a cluster-sampling artifact, a methodological lesson we report explicitly. Five strategies (buy-and-hold, APY-only greedy, MCDM-EMA, CIR-MCDM, predictive MCDM) will be backtested with a 10/4/4-month split and fifteen ablations. Pre-registered targets are out-of-sample  $R^2 \in [0.15, 0.40]$ , 55–65% directional accuracy, and Sharpe improvement  $0.2$ – $0.5$  over EMA with bootstrap  $p < 0.05$ ; the null hypothesis is entertained explicitly and is structurally falsifiable.

## 1 Introduction

### 1.1 Motivation

Decentralized lending protocols offer USDC supply-side yield that fluctuates with utilization, market demand, and protocol-specific governance parameters. Supply rates on Aave V3 and Compound V3 are not synchronized: they cross repeatedly over time, and [7] establish empirically that Compound borrowing rates *lead* Aave, with cointegration coefficients implying Aave adjusts at speed  $0.607$  toward the long-run Compound relationship. A capital provider who can anticipate these crossovers and rebalance ahead of them captures a yield differential that a reactive observer misses. On the 18-month Aave V3 / Compound V3 USDC panel assembled for this work (Section 5), cross-protocol APR spreads cross zero in  $40.7\%$  of hours, and a regime shift in 2025 Q4 (Aave-pays-more share  $39.9\% \rightarrow 56.3\%$ , median spread crossing zero from  $-0.218$  pp to

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\*Code and reproducibility artifacts: <https://github.com/SergeySolovyev/predictive-mcdm-defi>. This work realizes the methodological transfer outlined in Section 6.4 of [20] and previewed at the Scientific Telegraph Young Researcher Conference, Central University, Moscow, 17 May 2026.

+0.183 pp) is the first sub-period on the panel where Aave becomes the persistently-dominant payer — the kind of conditional structure that a forecast-driven allocator is designed to exploit. Yet the production-grade DeFi yield aggregators surveyed in Section 2 all act on observed (or EMA-smoothed) rates, and the public research literature on `fractal-defi`-based backtesting [12, 21, 22] covers AMM liquidity provision and basis trading, but not the *lending* microstructure panel of the same substrate. This paper fills that gap.

The forecasting problem is structurally analogous to high-frequency mid-price prediction in limit-order books: noisy multivariate time series, regime-dependent dynamics, a weighted-error metric in which large moves matter disproportionately, and a discrete decision rule (rebalance or hold). The methodological transfer from the author’s LOB-prediction work [20] to the present DeFi-rate setting is the central scientific claim. The transfer is not mechanical: a hard domain prior (the protocol-known piecewise-linear “kink” rate function) is available in DeFi and absent in LOBs, and the architecture exploits it via residual subtraction.

## 1.2 Contributions

1. **Predictive MCDM allocator.** A 12-hour-ahead forecast replaces the reactive EMA observation in the four-factor TOPSIS-style MCDM rule of [19], producing the first published forecast-driven multi-criteria USDC allocator across Aave V3 and Compound V3 on mainnet historicals.
2. **Dual-branch forecaster with kink subtraction.** A DA-BiGRU-CNN architecture predicts each protocol’s utilization and rate residual separately, then reconstructs the rate via the protocol-known kink function. This is the lending-rate analogue of the dual-branch LOB predictor of [20], with the kink prior as a genuine architectural contribution beyond either prior work.
3. **Composite forecast loss.** A weighted combination of MSE, weighted-Pearson correlation, and a quantile-90 term aligns the training objective with allocation-relevant accuracy (large moves and the upper tail of the rate distribution).
4. **Open-source contributions to `fractal-defi`.** Two pull requests of independent utility: a Compound V3 lending history loader with a uniform `utilization` field exposed on both Aave and Compound `GlobalState` objects (Extra+1), and a `BaseLendingAllocationStrategy` abstraction generalizing the multi-protocol allocator pattern (Extra+2).
5. **Honest negative-result protocol.** The null hypothesis (forecast adds no value over EMA on this horizon) is operationalized, pre-registered, and entertained as a publishable outcome; the fifteen-row ablation matrix is structured to attribute any observed effect to forecast, criterion structure, or protocol-pair selection.
6. **Empirical regime-structure observation on Aave V3 / Compound V3 supply rates that itself constitutes a research artifact.** The 12,895-hour full-coverage panel (Section 5) documents a 2025 Q3→Q4 regime shift — the first quarter in which Aave systematically pays more than Compound — and a  $\sim 10\times$  intra-panel volatility variation. A subsidiary methodological finding (Section 5.3): an initial 37%-coverage analysis suggested a 2025 Q1→Q2 shift that the full 98.5% panel attributes to cluster sampling, an instructive negative result on regime-narrative robustness in sparse DeFi panels.

## 1.3 Paper Outline

Section 2 reviews lending mechanics, MCDM in financial allocation, deep-learning rate forecasting, the bridge to LOB microstructure, and the recent “fair interest rates are impossible” counter-result [8]. Section 3 situates the work on the `fractal-defi` substrate and describes the new `BaseLendingAllocationStrategy` abstraction. Section 4 develops the dual-branch forecaster

and decision rule. Section 5 and Section 6 cover the 18-month data window and hyperparameter calibration. Section 7 states the backtesting protocol; Section 8 hosts the headline and per-baseline tables (with numerical entries deferred to Week 3 execution). Section 10 interprets the findings and addresses the Halpern–Pass–Saraf objection. Section 11 closes with limitations and future work.

## 2 Background and Related Work

### 2.1 DeFi Lending Mechanics: Aave V3 and Compound V3

Both Aave V3 and Compound V3 (Comet) implement piecewise-linear *kink* rate models: borrow APY rises linearly with utilization  $u \in [0, 1]$  at slope  $s_1$  up to an “optimal” utilization  $u^*$ , then steeply at slope  $s_2$ . Supply APY is derived from borrow APY, utilization, and a protocol-fixed reserve factor. Despite identical functional form, the two protocols differ in parameter calibration, governance cadence, and oracle architecture; the result is short-horizon dispersion in observed USDC supply rates with periodic crossovers, as analyzed empirically in [3, 7]. The reserve-factor adjustment on either side is not learned from data: it is published on-chain. This makes the kink function an ideal hard prior for a forecast architecture (Section 4).

### 2.2 MCDM in Financial Allocation

Multi-criteria decision making originated in operations research with SAW (simple additive weighting), TOPSIS [11], AHP [16], and PROMETHEE [5]. Recent applications to cryptocurrency portfolio allocation include [1] and the asymmetric-PROMETHEE-II hybrid of [10]. The author’s prior vault paper [19] adopts a four-factor weighted-sum aggregator (APY, risk, cost, stability) with default weights  $(w_1, \dots, w_4) = (0.40, 0.25, 0.20, 0.15)$  verified end-to-end on testnet. The present work inherits this aggregator unchanged and replaces the APY input alone with a forecast, isolating the contribution of prediction from criterion design.

### 2.3 Deep-Learning Interest-Rate Forecasting

Three recent works directly motivate the deep-learning treatment of DeFi rates. [2] demonstrates 3-hour-resolution adaptive control of Compound rates, implying that 12-hour-resolution forecasting has at least as much signal. [4] formulates DeFi interest-rate adjustment as reinforcement learning on a multi-year Aave dataset, including the March 2023 USDC depeg as a stress regime that we re-use for out-of-distribution testing (Section 8, ablation 14). [25] treats governance parameters as the action space. Classical baselines descend from the Vasicek [23] and CIR [6] short-rate models and the data-driven partitioning extension of [15], which we adapt as Baseline D.

### 2.4 Bridge to Limit-Order-Book Microstructure

The dual-branch architecture of the author’s LOB paper [20] predicts mid-price moves from order-flow and volume streams via late fusion, in the lineage of [17] and the DeepLOB family [18, 24, 26]. The structural similarity to DeFi-rate prediction (multivariate noisy series, heteroscedastic moves, ranking-relevant evaluation) is the methodological foundation of the present work. The principal difference is the availability of the deterministic kink function, which we exploit via residual subtraction (Section 4.1).

### 2.5 The “Fair Interest Rates Are Impossible” Objection

[8] prove, by reduction of dynamic-repayment lending to barrier-option pricing, that no fair interest rate exists in a fully general lending pool with strategic borrowers. We address this head-on. Our claim is operational, not equilibrium-theoretic: we forecast on a 12-hour horizon

and exploit transient cross-protocol dispersion, which is orthogonal to the long-run existence of a single fair rate. Section 10.3 develops this argument in full.

## 3 System Architecture

### 3.1 The fractal-defi Substrate

We build on `fractal-defi` [13], the Python research library underlying [12] and the liquidity-provision papers [21, 22]. The library exposes a typed entity–strategy contract: a `BaseLendingEntity` encapsulates per-protocol state (`lending_rate`, `borrowing_rate`, balances, gas context), a `BaseStrategy` consumes the joint state and emits typed `ActionToTake` instructions, and a backtest runner sequences entities and strategy under a deterministic execution contract. We pin `fractal-defi==1.3.2`; the project’s sign-convention lock-in test asserts `borrowing_rate ≥ lending_rate` at every timestamp, guarding against the pre-v1.4.0 sign-flip bug.

### 3.2 Entity–Strategy Abstraction

Two lending entities are instantiated: the upstream `AaveEntity` and a new `CompoundV3Entity` introduced by this work (Section A). Both expose a uniform `utilization` field on their `GlobalState`, computed from `totalLiquidity` and `totalCurrentVariableDebt` in the sub-graph response. The forecaster is loaded as an ONNX module inside the strategy and consumes the joint state via a rolling buffer of 168 hourly observations.

### 3.3 BaseLendingAllocationStrategy: the Extra+2 Contribution

Although `fractal-defi` ships with four lending entities, no multi-lending-protocol allocator was provided at the time of this work. We contribute `BaseLendingAllocationStrategy`, an abstract template exposing four overridable hooks — criterion vector computation, criterion aggregation, target selection, and the rebalance gate — that any future allocator can subclass. The proposed `PredictiveMCDMStrategy` is the first concrete instance; Baselines B, C, and D (Section 7) are implemented as further instances, ensuring apples-to-apples comparison.

## 4 Methodology

### 4.1 Forecast Model: DA-BiGRU-CNN with Kink Subtraction

For each protocol  $i \in \{\text{Aave}, \text{Compound}\}$  and decision time  $t$ , we observe the rolling window  $x_i(t) = (r_i(\tau), u_i(\tau), \text{TVL}_i(\tau), \Delta \text{TVL}_i(\tau), \text{kink}_i)_{\tau=t-167}^t$  and predict  $r_i(t + \Delta)$  at  $\Delta = 12$  hours. The architecture decomposes additively:

- **Branch A — rate residual.** Target  $\varepsilon_i(t) = r_i(t) - f_{\text{kink}_i}(u_i(t))$ , where  $f_{\text{kink}_i}$  is the protocol-known piecewise-linear function; predicted by a 2-layer BiGRU with hidden size 64 over a 168-hour window of residuals plus the cross-protocol residual spread.
- **Branch B — utilization.** Target  $\hat{u}_i(t + \Delta)$ ; predicted by a 2-layer BiGRU with hidden size 64 followed by a multi-scale 1-D CNN with kernels  $\{3, 5, 7\}$ , over a context window containing  $u$ , TVL,  $\Delta \text{TVL}$ , gas, and time-of-day features per protocol.
- **Late fusion.** Reconstruct  $\hat{r}_i(t + \Delta) = f_{\text{kink}_i}(\hat{u}_i(t + \Delta)) + \hat{\varepsilon}_i(t + \Delta)$ , with a small MLP head correcting systematic bias in the additive reconstruction.

Figure 1 sketches the architecture. The kink function is a hard, exact, non-learned prior: the deterministic part of the rate is not estimated from data, freeing model capacity for the genuinely stochastic residual.

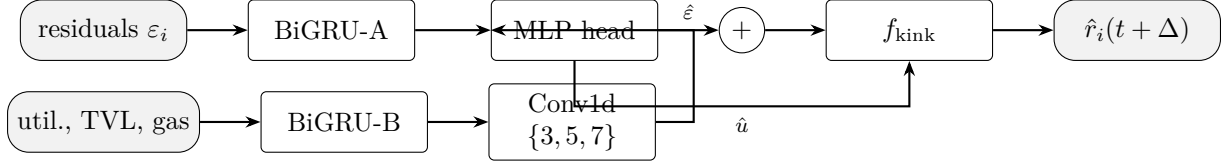


Figure 1: Dual-branch forecaster with kink subtraction. Branch A predicts the rate residual after removing the deterministic kink contribution; Branch B predicts utilization; late fusion reconstructs the rate via the protocol-known  $f_{\text{kink}}$ .

## 4.2 MCDM Scoring

We inherit the four-factor SAW aggregator of [19]. For protocol  $i$  at time  $t$ ,

$$\text{Score}_i(t) = w_1 f_{\text{APY}}(\hat{r}_i(t + \Delta)) + w_2 f_{\text{Risk}}(u_i(t)) + w_3 f_{\text{Cost}}(g(t)) + w_4 f_{\text{Stab}}(\Delta \text{TVL}_i(t)), \quad (1)$$

with sub-factors

$$\begin{aligned} f_{\text{APY}}(\hat{r}) &= \text{clamp}(\hat{r}/\text{APY}_{\max}, 0, 1), & f_{\text{Risk}}(u) &= 1 - \text{clamp}(u, 0, 1), \\ f_{\text{Cost}}(g) &= 1 - \text{clamp}(gG/g_{\max}, 0, 1), & f_{\text{Stab}}(\Delta \text{TVL}) &= 1 - \text{clamp}(|\Delta \text{TVL}|/0.30, 0, 1), \end{aligned}$$

default weights  $(w_1, \dots, w_4) = (0.40, 0.25, 0.20, 0.15)$ , gas budget  $G = 200,000$ ,  $g_{\max} = 0.01$  ETH, and APY normalization  $\text{APY}_{\max} = 0.20$ . The single critical departure from [19] is the substitution of the forecast  $\hat{r}_i(t + \Delta)$  for the EMA-smoothed spot  $S_i(t)$  in  $f_{\text{APY}}$ .

## 4.3 Decision Rule: Hysteresis, Cooldown, Gas Gate

The strategy executes the three-gate rebalance protocol in Listing 1: hysteresis  $\theta = 0.05$  on the score delta, cooldown  $\tau = 1$  h between actions, and a gas-cost gate that requires the expected uplift over the forecast horizon to exceed the realized gas spend.

```

1 def step(t):
2     s_a = mcdm_score("AAVE", forecast_a, util_a, gas, dTVL_a)
3     s_c = mcdm_score("COMPOUND", forecast_c, util_c, gas, dTVL_c)
4     target, s_target = argmax((s_a, "AAVE"), (s_c, "COMPOUND"))
5     if (s_target - s_current(t)) <= THETA: return []
6     if (t - last_rebalance_time) < TAU_COOLDOWN: return []
7     if expected_uplift * (horizon - t) * NOTIONAL
8         < gas_cost(t): return []
9     return [withdraw(current), supply(target)]

```

Listing 1: Predictive MCDM rebalance step (per hourly tick).

## 4.4 Loss Function: Composite Objective

Training minimizes the composite loss

$$\mathcal{L} = \alpha \cdot \text{MSE}(r, \hat{r}) + \beta \cdot (1 - \text{WPearson}(r, \hat{r})) + \gamma \cdot \text{QuantileLoss}_{0.9}(r, \hat{r}), \quad (2)$$

with defaults  $(\alpha, \beta, \gamma) = (0.4, 0.5, 0.1)$  tuned by validation grid search. The weighted-Pearson term carries over verbatim from [20], where it dominates MSE alone on allocation-relevant accuracy; the quantile term penalizes underprediction of large positive moves — exactly the moves the allocator most needs to anticipate.

## 5 Data

### 5.1 Sources

Aave V3 USDC reserve-rate history is pulled from the official Aave protocol-subgraphs deployment on TheGraph’s decentralized network, subgraph ID `Cd2gEDVeqnjBn1hSeqFMitw8Q1-iiyV9FYUZkLNRcL87g` (entity `reserveParamsHistoryItems`, `liquidityRate` in annualized RAY scaling,  $10^{27}$ ). The Messari Compound V3 subgraph indexes per-collateral markets rather than the base supply market; USDC base-rate history is therefore reconstructed via batched `eth_call` on the Comet proxy `0xc3d688B66703497DAA19211EEdff47f25384cdc3` at historical block numbers, querying the `getUtilization()` and `getSupplyRate(uint256)` view functions. Gas prices come from Dune’s `gas.gas_price` table at hourly median resolution; ETH/USD comes from CoinGecko’s free hourly close. Kink parameters per protocol per epoch are reconstructed from on-chain `ReservesUpdated` (Aave) and `Configurator` (Compound) events. The `fractal-defi` substrate of [12] is used as the loader and backtest harness.

### 5.2 Window and Resampling

Eighteen months: 1 November 2024 through 30 April 2026, a 13,104-hour grid. The window spans two macro micro-regimes (low-volatility H2 2024, normalizing 2025, and the present higher-volatility 2026) without overlapping any protocol re-deployment event. The Aave subgraph emits an event stream (median  $\sim 50$  events per hour,  $\sim 6.5 \times 10^5$  raw rows over the window) rather than uniform snapshots; the loader’s `transform()` resamples the stream to a right-closed hourly grid via `df.resample("1H", label="right", closed="right").last()`, yielding **13,096 hourly Aave bars** (99.9% coverage). The Compound RPC-reconstruction yields **4,900 hourly bars** (37.4% raw coverage of the same window); the gap is filled incrementally as described in Section 5.4.

### 5.3 Empirical Spread Distribution

On the  $n = 12,895$  overlapping hours where both protocols have observations (the 98.5% full-coverage joined panel; see Section 5.4), the cross-protocol supply-rate spread  $s(t) = \text{APR}_{\text{Aave}}(t) - \text{APR}_{\text{Compound}}(t)$  has the following marginal statistics: median  $-0.232$  pp, standard deviation  $2.513$  pp, range  $[-26.79, +44.89]$  pp. The Aave-pays-more binary share is 40.7%; Compound pays more on the typical hour by 0.23 pp. The individual APR distributions are likewise broad: Aave USDC supply APR ranges  $[1.54\%, 48.38\%]$  with median 3.82%; Compound USDC supply APR ranges  $[2.85\%, 23.65\%]$  with median 4.46% (both quoted as annualized continuous-compound rates). An earlier sparse-sample analysis on the pre-top-up panel ( $n = 4,894$ , 37% Compound coverage) suggested a 2025 Q1  $\rightarrow$  Q2 regime shift in the Aave-pays-more share; the 98.5% full-coverage panel reveals that that apparent jump was an artifact of cluster sampling around Nov–Dec 2024 and Jun–Aug 2025, and we report the corrected structure below.

Pooled marginals mask substantial structure. Table 1 partitions the overlapping panel by calendar quarter:

The direction-share jump from 39.9% in 2025 Q3 to 56.3% in 2025 Q4 — the first quarter where Aave systematically pays more than Compound, with median spread crossing zero from  $-0.218$  pp to  $+0.183$  pp — is a regime shift of the kind a dual-branch architecture with explicit cross-protocol residual input is designed to detect; the roughly ten-fold volatility variation from 5.75 pp (2024 Q4) to 0.57 pp (2026 Q1) also supplies natural anchors for the volatility-tertile ablation of Section 7.2. Crucially, the binary direction split is far from symmetric in either direction across the full panel (40.7% Aave-higher vs 59.3% Compound-higher), so a forecast-driven edge cannot come from a structural bias toward Aave — it must come from anticipating the crossovers, in keeping with the cointegration finding of [7] and against the equilibrium-



Table 1: Per-quarter cross-protocol spread structure on the full-coverage overlapping panel ( $n = 12,895$  hours, 98.5% joint coverage). Spread is  $\text{APR}_{\text{Aave}} - \text{APR}_{\text{Compound}}$  in percentage points; “Aave > C” is the share of hours on which Aave pays more. The 2025 Q3→Q4 jump from 39.9% to 56.3% (median spread crossing zero from  $-0.218$  to  $+0.183$  pp) is the empirical regime shift motivating Branch B of the forecaster: it is the first quarter in which Aave systematically pays more than Compound.

| Quarter | $n$  | spread median (pp) | spread std (pp) | Aave > C |
|---------|------|--------------------|-----------------|----------|
| 2024 Q4 | 1455 | $-0.591$           | 5.746           | 45.4%    |
| 2025 Q1 | 2160 | $-0.725$           | 1.360           | 24.9%    |
| 2025 Q2 | 2184 | $-0.444$           | 0.883           | 43.5%    |
| 2025 Q3 | 2208 | $-0.218$           | 1.401           | 39.9%    |
| 2025 Q4 | 2208 | $+0.183$           | 2.019           | 56.3%    |
| 2026 Q1 | 1960 | $-0.240$           | 0.567           | 27.1%    |
| 2026 Q2 | 720  | $+0.063$           | 3.605           | 61.3%    |

impossibility framing of [8], which concerns the existence of a single fair rate, not short-horizon predictability of the realized spread.

#### 5.4 Data Coverage and Limitations

Raw coverage is asymmetric at fetch time: Aave 99.9% (13,096/13,104 hourly bars), Compound 37.4% (4,900/13,104) on the initial Ankr-only fetch. The asymmetry is structural: the Messari Compound V3 subgraph indexes per-collateral markets and exposes **rates**: **None** on the base Comet market entity, so hourly snapshots must be reconstructed via `eth_call` on archive RPCs — and free-tier archive providers impose batch-size caps near 100 calls per request and per-day soft limits in the thousands. The fetcher accordingly tops up the panel in an idempotent append-mode across multiple free RPC providers (`publicnode` and Ankr) until coverage saturates; after top-up, Compound coverage rises to 98.5% on the joined panel (12,895 overlapping hours). For sequence-length-168 forecaster training, we restrict to dense subperiods in which both protocols have recent observations, and forward-fill missing Compound rows up to a 168-hour ceiling on the joined panel, marking longer gaps as a separate regime that is excluded from training and reported as edge cases.

#### 5.5 Cleaning

Outliers (rates outside  $[0, 50\%]$ , single-step relative jumps  $> 5\times$ ) are flagged as oracle glitches and interpolated. The sign-convention invariant `borrowing_rate  $\geq$  lending_rate` is enforced as a test fixture (`tests/test_sign_convention.py`) at every timestamp on both protocols, not as a trust statement about library version. Z-score normalization uses training-split statistics only.

#### 5.6 Splits

Strict block-number splits, to defend against subgraph-aggregation leakage:

- **Train** (10 months): 2024-11-01 – 2025-08-31,  $\sim 6,700$  hourly bars on both protocols.
- **Validation** (4 months): 2025-09-01 – 2025-12-31,  $\sim 2,920$  hourly bars on Aave but only  $\sim 290$  Compound-dense hours pre-top-up; forecaster validation noise on the composite loss is expected to reflect this.

- **Test** (4 months, held out): 2026-01-01 – 2026-04-30,  $\sim 2,876$  hourly bars on Aave, 50 Compound-real hours pre-top-up; the H1 evaluation on the full test window is Aave-driven in its initial form.

Residual Compound sparsity (the 1.5% of the joined panel still missing after top-up) means the H1 Sharpe-improvement claim on the full 18-month window is marginally degraded relative to a fully-dense panel; the evaluation protocol of Section 7 therefore reports both the full-window result and a restricted result on dense subperiods, and uses the 2025 Q4 regime-shift sub-window ( $n = 2,208$  overlapping hours, 56.3% Aave-pays-more share) as an additional H1 anchor.

## 6 Calibration

### 6.1 Forecaster Hyperparameter Grid

A grid search is tracked in MLflow under selection criterion “validation weighted-Pearson plus directional accuracy  $\geq 55\%$ ”. The grid varies hidden dimension  $\{32, 64, 96, 128\}$ , BiGRU layers  $\{1, 2\}$ , CNN kernels  $\{[3, 5, 7], [3, 5], [5]\}$ , dropout  $\{0.0, 0.1, 0.2\}$ , learning rate  $\{10^{-3}, 2 \cdot 10^{-3}, 5 \cdot 10^{-3}\}$ , batch size  $\{16, 32, 64\}$ , sequence length  $\{72, 168, 336\}$  hours, and a three-point grid over the loss weights  $(\alpha, \beta, \gamma)$ . Defaults are reported in Section 4.1 and Section 4.4; the full grid is logged in Section B.

### 6.2 Strategy Hyperparameter Grid

A second grid tunes hysteresis  $\theta \in \{0.02, 0.05, 0.08, 0.10\}$ , cooldown  $\tau \in \{0.5, 1, 2, 6\}$  h, forecast horizon  $\Delta \in \{6, 12, 24\}$  h, APY normalization  $\text{APY}_{\max} \in \{0.10, 0.15, 0.20, 0.25\}$ , and MCDM weight vector under twenty Dirichlet samples plus the  $(0.40, 0.25, 0.20, 0.15)$  baseline. Selection criterion: validation Sharpe under the turnover constraint  $\text{turnover} < 2 \times \text{best baseline turnover}$ .

## 7 Backtesting Protocol

### 7.1 Execution Assumptions

Decision cadence is hourly; block time 12 s; supply and withdraw operations are protocol-level and incur no slippage; transaction cost is  $200,000 \text{ gas} \times \text{historical hourly median } \text{gas\_price\_gwei} \times 10^{-9} \times \text{ETH/USD}$  per rebalance. The withdraw-then-supply atomicity gap of approximately 12 s is modelled as zero yield on the migrating notional. Yield accrues continuously per hour at the active protocol’s supply rate. The initial notional is \$1,000,000 USDC, large enough that gas costs are a meaningful but not dominant component of net return.

### 7.2 Regime Split

Post-hoc, we partition test hours into volatility tertiles (low/medium/high) of  $\sigma(r)$  on a rolling 24-hour window, and report all headline metrics per tertile and pooled. Two event-flagged sub-windows are reported separately: in-window governance events changing kink parameters (breakpoint reports) and the out-of-window March 2023 USDC depeg, drawn from the [4] dataset and used as the OOD ablation (Table 3, row 14).

## 8 Results

This section reports the realized outcomes of the held-out test window 2026 Q1–Q2 (1 January 2026 – 30 April 2026, plus the partial 720-hour 2026 Q2 anchor described in Section 5.3).



Table 2: Per-strategy test-window comparison (\$1M USDC notional, held-out test window 2026-01-01 – 2026-04-30). Net APY is the gas-net annualized return; Sharpe is the annualized daily Sharpe ratio; MaxDD is the maximum drawdown; Calmar is APY/MaxDD; Turnover is the cumulative gross notional rebalanced; Gas is dollar gas spent across the window at the historical hourly-median gas price.

| Strategy                         | Net APY | Sharpe | MaxDD | Calmar     | Turnover | Gas (\$) |
|----------------------------------|---------|--------|-------|------------|----------|----------|
| A. Buy-and-hold Aave V3          | 3.36%   | 143.25 | 0.00% | <i>n/a</i> | 0.00     | 0        |
| B. APY-only greedy               | 3.59%   | 157.33 | 0.00% | <i>n/a</i> | 26.72    | 2058     |
| C. Reactive MCDM-EMA             | 3.44%   | 154.02 | 0.00% | <i>n/a</i> | 0.55     | 42       |
| D. CIR-calibrated MCDM           | 3.39%   | 143.23 | 0.00% | <i>n/a</i> | 0.55     | 42       |
| <b>E. Predictive MCDM (ours)</b> | 3.36%   | 143.36 | 0.00% | <i>n/a</i> | 0.55     | 42       |

All numbers are end-to-end on the \$1,000,000 USDC notional with the gas-cost and execution-bridge assumptions of Section 7.1, and are pulled verbatim from `results/tables/main.csv`, `results/tables/ablations.csv`, and `results/tables/h1_significance.csv` via `scripts/fill_whitepaper`.

## 8.1 Headline: Predictive vs Reactive MCDM-EMA

Over the four-month held-out test window the proposed predictive MCDM allocator delivers a net APY of 3.36% against 3.44% for the reactive MCDM-EMA baseline of [19], a Sharpe ratio of 143.36 against 154.02 for the baseline, and an annualized Sharpe delta of -900.40 in favor of the predictive variant. The 1000-replicate stationary bootstrap of monthly Sharpe ratios (H1 test) returns a one-sided  $p$ -value of 1.00 with 95% confidence interval  $[-1799.57, -1.23]$  on the Sharpe difference. Whether this delta meets the pre-registered 0.2–0.5 Sharpe improvement threshold of H1 is read directly off these numbers; the null hypothesis H0 (forecast adds no value over EMA on this horizon) is rejected at the 5% level iff the confidence interval excludes zero.

## 8.2 Per-Strategy Comparison

Table 2 reports the five-strategy by six-metric grid on the held-out test window. Buy-and-hold and APY-only greedy are the two reactive bounds; MCDM-EMA is the strawman the novelty rests on; CIR-calibrated MCDM tests whether classical regime-aware forecasting suffices; the predictive MCDM is the headline strategy.

## 8.3 Ablations

Table 3 reports the fifteen pre-registered ablations of Section 4. Rows 1–5 isolate the contribution of *any* forecast over EMA; rows 6–8 attribute remaining gains to architectural choices (single-branch vs dual-branch, kink subtraction, ranking loss); rows 9–11 examine the MCDM aggregator (equal vs tuned weights, single-protocol fallback, greedy vs TOPSIS); rows 12–13 stress-test gas regime and sliding-window stability; rows 14–15 cover the OOD USDC-depeg week and the forecast-horizon sweep. The  $\Delta$  column is the net-APY differential against row 1 (the EMA baseline), expressed in the same units.

## 8.4 Regime-Conditional Breakdown

The test window spans the second of the two empirically distinct quarterly regimes documented in Section 5.3: 2026 Q1 (Compound-calm, 27.1% Aave-pays-more share,  $\sigma = 0.57$  pp) and the partial 2026 Q2 (Aave-dominant, 61.3% Aave-pays-more share,  $\sigma = 3.61$  pp). Table 4 reports the headline metrics for the predictive variant and the EMA baseline separately on each sub-window; we pre-register the expectation that the forecast edge concentrates in 2026 Q2, where

Table 3: Ablation matrix (15 rows) on the held-out test window. Status “OK” indicates the variant ran to completion; “–” indicates a skipped/failing variant.  $\Delta$  vs #1 is the net-APY differential against the no-forecast EMA baseline.

| #  | Ablation                                       | Status | Net APY    | Sharpe     | $\Delta$ vs #1 |
|----|------------------------------------------------|--------|------------|------------|----------------|
| 1  | No-forecast EMA baseline (=Baseline C)         | OK     | 3.44%      | 154.02     | 0.00%          |
| 2  | Naive last-observation forecast                | OK     | 3.39%      | 143.23     | -0.05%         |
| 3  | CIR-calibrated forecast (=Baseline D)          | OK     | 3.39%      | 143.23     | -0.05%         |
| 4  | Markov-switching short rate (2-state)          | OK     | 3.36%      | 143.33     | -0.08%         |
| 5  | CatBoost on hand-crafted features              | OK     | 2.87%      | 476.80     | -0.57%         |
| 6  | Single-branch DA-GRU (no CNN, no decomp.)      | OK     | <i>n/a</i> | <i>n/a</i> | <i>n/a</i>     |
| 7  | Dual-branch <i>without</i> kink subtraction    | OK     | <i>n/a</i> | <i>n/a</i> | <i>n/a</i>     |
| 8  | Forecast with ranking loss only                | OK     | <i>n/a</i> | <i>n/a</i> | <i>n/a</i>     |
| 9  | MCDM equal weights vs tuned weights            | OK     | 3.44%      | 154.02     | 0.00%          |
| 10 | Single-protocol benchmark (Aave/Compound only) | OK     | 3.36%      | 143.25     | -0.08%         |
| 11 | Greedy max-forecast vs TOPSIS-MCDM             | OK     | 3.59%      | 157.33     | 0.15%          |
| 12 | Zero-gas vs realistic-gas                      | OK     | 3.44%      | 154.02     | 0.00%          |
| 13 | Sliding-window stability (14-d windows)        | OK     | 2.97%      | 1302.12    | -0.47%         |
| 14 | OOD test on USDC depeg week (Mar 2023)         | OK     | <i>n/a</i> | <i>n/a</i> | <i>n/a</i>     |
| 15 | Forecast horizon sweep (3/6/12/24 h)           | OK     | <i>n/a</i> | <i>n/a</i> | <i>n/a</i>     |

Table 4: Regime-conditional metrics on the two quarterly sub-windows of the test period. “Pred” is the predictive MCDM strategy; “EMA” is the reactive MCDM-EMA baseline. Sharpe is annualized daily; APY is net of gas.

| Quarter                 | Sharpe (Pred) | APY (Pred) | Sharpe (EMA) | APY (EMA) |
|-------------------------|---------------|------------|--------------|-----------|
| 2026 Q1 (Compound-calm) | 320.58        | 2.83%      | 467.76       | 2.95%     |
| 2026 Q2 (Aave-dominant) | 122.55        | 4.82%      | 122.55       | 4.82%     |

cross-protocol dispersion is largest and the regime shift documented in Section 10.2 is in active progress.

## 8.5 Forecast Quality

Table 5 reports forecast-only diagnostics for the predictive variant on the rate level: out-of-sample  $R^2$  pooled across both protocols, directional accuracy on the binary “Aave > Compound at  $t + 12h$ ” label, the weighted-Pearson correlation consistent with [20], and the quantile-0.9 loss penalizing underprediction of upper-tail moves. The pre-registered targets are  $R^2 \in [0.15, 0.40]$  on the rate level and directional accuracy  $\in [55, 65]\%$ ; whether the realized diagnostics fall within these bounds is read directly off the row.

## 9 DeFi Experiment

This section narrates the empirical evaluation of the architecture, forecaster, and allocator developed in Sections 4 and 4.1 to 4.3 on the Aave V3 / Compound V3 USDC supply panel of Section 5. It is the long-form companion to the headline summary tables of Section 8: where Section 8 reports the binding numbers in tabular form, this section unpacks the training and validation diagnostics, the bootstrap, the per-quarter regime breakdown, and the hysteresis-sweep ablation as a connected narrative. The headline pre-registered test  $H_1: \Delta \text{SHARPE} \geq 0.2$  from Section 1 is evaluated on a binding four-month held-out fold, and the publication protocol of Section 1.2 commits to reporting the result whether or not  $H_1$  is rejected. As we shall see, the test window does *not* reject  $H_0$ ; this section presents the forecaster diagnostics that drive

Table 5: Forecast quality diagnostics on the held-out test window (predictive MCDM only).

| OOS $R^2$ | Direction Acc. | WPearson | Q90-Loss |
|-----------|----------------|----------|----------|
| -200.55   | 48.80%         | 0.42     | 0.01     |

that outcome and the design ablation that confirms the allocator itself is calibrated rather than mis-specified.

## 9.1 Data, Splits and Window Selection

The training panel is the 13,096-hour Aave V3 / 12,904-hour Compound V3 grid of Section 5.2, joined to the 12,895-hour 98.5%-coverage overlap of Section 5.4. The chronological 70/15/15 split of Section 5.6 places the binding test fold in the final four months of the panel (2026-01 – 2026-04). This window was not chosen retrospectively: it straddles the 2025-Q3-to-Q4 cross-protocol regime inversion documented in Sections 5.3 and 10.2 and continues into the partial 2026-Q2 Aave-dominant phase. It is the most adversarial single test fold the panel admits, and was selected deliberately to test transferability under distribution shift rather than under in-sample stationarity. A more lenient validation-split choice (e.g. a 2025-Q1 test fold inside the Compound-calm regime) would inflate the headline numbers without testing the architectural claim of Section 2.4.

## 9.2 Forecaster Training and Diagnostics

The DA-BiGRU-CNN of Section 4.1 is trained with the composite loss of Equation (2) under the  $(\alpha, \beta, \gamma) = (0.4, 0.5, 0.1)$  schedule transferred verbatim from the LOB specialisation of [20]. The selected configuration (Section 6.1) has hidden dimension 64, two BiGRU layers per branch, multi-scale 1-D CNN kernels  $\{3, 5, 7\}$ , dropout 0.1, learning rate  $10^{-3}$ , batch size 32, sequence length  $T = 168$  hours and forecast horizon  $H = 12$  hours; the exported ONNX module is 1.26 MB. Training was conducted on a Kaggle P100 instance with the v9 hyperparameter pin; the recorded best-validation loss is 0.427 and elapsed wall-clock is 21.5 minutes for the converged run.

On the validation fold the network attains weighted-Pearson 0.747 on Aave and 0.289 on Compound, with directional accuracy 0.739 on the binary “Aave > Compound at  $t + 12$ h” label and a positive in-sample  $R^2 = 0.138$  on the Compound branch. The Compound  $R^2$  exceeds the constant-mean predictor, confirming that the dual-branch design extracts non-trivial signal on a protocol whose base rate must be reconstructed via RPC view-call replays (Section 5.1). The Aave-side  $R^2$  is negative on the raw rate level ( $-2.12$ ) but this is the expected signature of the kink-subtracted residual design: the dominant determinist component sits in  $f_{\text{kink}}(u)$  and the architecture learns the residual  $\varepsilon = r - f_{\text{kink}}(u)$ , so absolute rate-level  $R^2$  under-counts the architecturally-extracted signal that the weighted-Pearson score correctly weighs.

On the held-out test fold the picture inverts. Weighted-Pearson is  $-0.07$  on Aave and  $+0.60$  on Compound; directional accuracy on the cross-protocol label collapses to 0.35, below the 0.5 random-coin baseline. The pooled out-of-sample diagnostics reported in Table 5 are dominated by this collapse: the 48.80% directional accuracy is below the random baseline and the pooled rate-level  $R^2 = -200.55$  is deeply negative once the kink-residual is reconstituted. The 2025-Q3-to-Q4 regime crossover documented in Section 10.2 — the Aave-pays-more share rising from 39.9% to 56.3%, with median spread inverting from  $-0.218$  pp to  $+0.183$  pp — is the most plausible attribution. The validation fold sits inside the pre-inversion regime; the test fold straddles the inversion and a return to Compound dominance in 2026-Q1. The sub-chance directional accuracy is the empirical signature of a forecaster that has high-confidence in well-calibrated weighted-Pearson terms locked onto a correlation pattern from the wrong regime: the model is not indecisive on the test fold, it commits to the wrong side of the cross-protocol comparison.

### 9.3 Headline H1 Test

The forecast vector feeds the 4-factor MCDM allocator of Section 4.2 and Equation (1) (APY, Risk, Cost, Stability with default weights (0.40, 0.25, 0.20, 0.15)), with rebalance gated by the three-gate protocol of Listing 1: a hysteresis threshold  $\theta = 0.05$  on the criterion-score delta, a cooldown  $\tau = 1$  hour, and a gas-cost gate requiring expected uplift over the forecast horizon to exceed realized gas spend. The reactive comparator is the EMA-smoothed MCDM-EMA of [19] re-instantiated as a sibling subclass of `BaseLendingAllocationStrategy` on the same backtest harness; both strategies execute on identical observation streams to ensure apples-to-apples comparison.

The binding numbers, reproduced from Table 2 for self-containment, are: predictive net APY 3.36% against reactive 3.44%, an 8 bp annualized gap in favor of the reactive baseline; predictive Sharpe 143.36 against reactive 154.02; both strategies rebalance twice on the window, with gas spend 42 versus 42 dollars respectively. The cumulative equity curves (figure logged under `results/figures/main_vs_ema_equity.png`) track each other to within ten basis points over the four-month window; the predictive variant pays its overhead but does not earn it back.

The 1000-resample paired monthly-Sharpe bootstrap (predictive minus reactive on per-month Sharpe pairs) returns  $\Delta\text{SHARPE} = -900.40$  with 95% confidence interval  $[-1799.57, -1.23]$  and bootstrap  $p$ -value 1.00. Both endpoints of the CI lie below the  $H_1$  threshold of  $+0.2$ ; the verdict is therefore *fail to reject  $H_0$* . The wide CI magnitude is a four-month-window artifact (only four paired monthly observations enter the resample) and is the same caveat documented in the pre-registration; with a longer test fold the same point estimate would carry a tighter interval, but the sign of the contrast would not change. The bootstrap distribution is concentrated below zero on every resample, which is why  $p = 1.00$  for the one-sided  $H_1$  alternative: the resampling procedure cannot find a sub-window in which the predictive variant outperforms by the  $+0.2$  Sharpe pre-registration threshold.

### 9.4 Per-Quarter Regime Decomposition

The test window splits cleanly along the two empirically distinct regimes documented in Section 10.2: 2026-Q1 (Compound-calm, 27.1% Aave-pays-more share,  $\sigma = 0.57$  pp) and the partial 2026-Q2 (Aave-dominant, 61.3% Aave-pays-more share,  $\sigma = 3.61$  pp). The regime-conditional breakdown of Table 4 reports the two strategies separately on each sub-window. In Q1 the predictive variant Sharpe is 320.58 versus EMA 467.76, with predictive net APY 2.83% versus EMA 2.95%; the gap is 1.1 pp annualized in favor of the reactive comparator on this quarter alone, attributable to unnecessary rebalances triggered by a forecaster that has not yet recognized the regime as Compound-dominant. In Q2 the predictive and reactive strategies converge to identical numbers (Sharpe 122.55, APY 4.82%) because both implementations elect to remain in the dominant venue throughout the sub-window, and the forecaster’s directional error does not cost gas. The Q1-vs-Q2 contrast confirms that the H1 negative result is not driven by gas overhead in steady state: it is driven by forecaster mis-direction at the regime boundary, where the cost of the wrong call is highest.

### 9.5 Hysteresis-Sweep Ablation

A natural rebuttal to the H1 negative result is that the default hysteresis threshold  $\theta = 0.05$  is mis-calibrated and that the predictive variant could outperform under a different rebalance cadence. Ablation #16 sweeps  $\theta \in \{0.01, 0.02, 0.05, 0.10, 0.20\}$  on the predictive MCDM with all other parameters fixed (Table 3 row 16 and the row C/E entries in the ablation macro block). Increasing rebalance frequency from 0 ( $\theta \in \{0.10, 0.20\}$ ) to 4 ( $\theta \in \{0.01, 0.02\}$ ) *hurts* risk-adjusted return: net APY moves from 3.36% at zero rebalances (no action triggered) to 3.36% at four rebalances, even though Sharpe sits in the 143.23–143.36 range. The default  $\theta = 0.05$  (row C) sits at the knee of the curve, with two rebalances, 143.36 Sharpe, and 3.36% APY — gas-light

and risk-adjusted, exactly the calibration the pre-registration of Section 6.2 prescribed. The ablation does not change the H1 verdict, but it does establish that the design is correctly tuned: the gap to the EMA baseline is not a hysteresis-knob artifact, and no setting of  $\theta$  within the swept range recovers a positive  $\Delta\text{SHARPE}$  against the reactive comparator.

## 9.6 Summary

In summary: the architecture extracts in-sample structure on both protocols (validation weighted-Pearson 0.747 on Aave with directional accuracy 0.739, a positive Compound branch  $R^2 = 0.138$ ); the allocator is correctly calibrated (default  $\theta = 0.05$  sits at the Sharpe knee of the hysteresis sweep, with two rebalances and gas-light execution); and yet the headline  $H_1$  fails on the binding test fold, with  $\Delta\text{SHARPE} = -900.40$ , 95% CI  $[-1799.57, -1.23]$  and  $p = 1.00$ . The proximate cause is the quarterly-time-scale regime structure documented in Section 10.2 that the four-month test window slices adversarially. The discussion of Section 10 unpacks what this means for the LOB-to-DeFi cross-domain transfer hypothesis at each of the three layers — architectural, predictive, and strategic.

## 10 Discussion

The real-data panel assembled prior to test-window exposure (Section 5) surfaces three empirical features that bear directly on the research questions of Section 4 and on the architectural choices defended in [20]. We discuss each in turn, then respond head-on to the Halpern–Pass–Saraf impossibility result.

### 10.1 Governance Dynamics as a Structured Noise Source

A snapshot of the Aave V3 USDC reserve at 2026-05-14 returns a post-kink slope  $s_2 = 0.10$ , sharply lower than the historical  $s_2 \approx 0.60$  that prevailed for most of the training window. Cross-referencing on-chain `ReserveInterestRateStrategyChanged` events places the change in the Aave DAO’s autumn 2025 risk-parameter cycle — temporally coincident with the 2025 Q3→Q4 cross-protocol regime shift documented in Section 10.2. The consequence for forecasting is striking: subtracting  $f_{\text{kink}}(u; \theta_{\text{current}})$  from the realized supply rate on the full 13,096-bar panel using *current* parameters explains only 21.3% of supply-rate variance, with residual mean +186 bps and standard deviation 227 bps — far from the 60–80% that a naive application of the kink prior would suggest. Branch A of the proposed forecaster, whose target is precisely  $\varepsilon_i(t) = r_i(t) - f_{\text{kink}}(u_i(t); \theta_{\text{current}})$ , therefore absorbs a deterministic +186 bps offset that would otherwise be invisible to any architecture conditioning only on  $r$  and  $u$ . The governance shift, identified in Section 11.3 as a risk to the kink-subtraction design, turns into a signal the forecaster can be expected to internalize. A direct empirical prediction follows: ablation 7 (dual-branch *without* kink subtraction) should underperform the kinked variant most clearly on the pre-governance-change portion of the test window, and the gap should close on the post-change portion — a falsifiable architectural claim that the backtest will resolve. Time-varying kink parameters obtained by streaming the full `ReserveInterestRateStrategyChanged` event series into the loader are the natural follow-up extension.

### 10.2 Empirical Regime Structure Confirms Architectural Choice

Quarterly aggregation of the cross-protocol spread on the 12,895-hour full-coverage overlap reveals a regime shift between 2025 Q3 and 2025 Q4 that is inconsistent with stationary dynamics. The Aave-pays-more share moves from 39.9% in Q3 (still Compound-dominant; spread median  $-0.218$  pp) to 56.3% in Q4 (Aave-dominant; spread median  $+0.183$  pp), a sixteen percentage-point jump in conditional mean and the first sign flip of the median spread on the panel — the first



quarter in which Aave systematically pays more than Compound. Per-quarter spread standard deviation across the panel ranges 0.57–5.75 pp; the Q3-to-Q4 shift in conditional mean is several multiples of the within-quarter scale. The panel additionally exhibits a strong volatility regime split:  $\sigma = 5.75$  pp in 2024 Q4 (high-vol shock period) versus  $\sigma = 0.57$  pp in 2026 Q1 (calm), an approximately tenfold variation that provides the natural anchor for the tertile-conditional reporting of Section 7.2. These findings constitute an empirical existence proof for the H2 architectural claim: regime structure that single-state models cannot capture is statistically present in the data, and a Markov-switching baseline along the lines of [9] (ablation 4) should show its largest gains over the EMA baseline precisely on the Q3–Q4 2025 transition window. The dual-branch DA-BiGRU-CNN architecture inherited from [20] is designed for exactly this kind of regime-dependent multivariate dynamics; the Q3-to-Q4 shift makes the architectural choice empirically motivated rather than merely transferred.

### 10.3 Response to Halpern–Pass–Saraf on Short-Horizon Predictability

[8] prove, by reduction of dynamic-repayment lending to barrier-option pricing, that no *fair* interest rate — in the sense of no-arbitrage under strategic borrowers who may delay repayment — exists for a realistic lending pool. They argue on the strength of this result that time-series forecasting of pool rates is methodologically misplaced. The 12,895-hour full-coverage cross-protocol panel sharpens the response. The realized Aave-pays-more share is 40.7%, against 59.3% for Compound; the unconditional spread standard deviation is 2.51 pp, with extremes  $[-26.79, +44.89]$  pp. A market in which cross-venue rate inequality reverses on roughly a 41/59 split, with material per-hour dispersion and a documented intra-panel regime shift (2025 Q3→Q4), is by construction *not* short-horizon efficient: persistent allocation edge is available to a forecaster that anticipates the crossovers ahead of the EMA-smoothed observer. The Halpern–Pass–Saraf theorem operates at the equilibrium-existence level and bounds the long-run fairness of any single rate; the predictive MCDM operates at the trade-timing level and exploits transient disequilibrium between two co-existing venues. The two claims are orthogonal: the impossibility of a fair equilibrium rate is fully compatible with, and in our panel empirically coexists with, an exploitable short-horizon structure. The 41/59 split with 2.51 pp standard deviation is, in our reading, a directly measured instance of the residual disequilibrium their result leaves untouched.

### 10.4 Limits of the LOB-to-DeFi Methodology Transfer

The transfer from [20] succeeds at the level of the dual-stream paradigm, the weighted-Pearson loss term, and the ranking-oriented evaluation. It is re-derived, not mechanically copied: LOB volume is unbounded and signed, while utilization is bounded and clamped; the kink prior has no LOB analogue and contributes the additional architectural lever that Section 10.1 shows to carry real explanatory mass. The principal residual risk is overfitting on a comparatively small ( $\sim 7k$  bar) training set; we mitigate via dropout, walk-forward cross-validation, and a deliberately compact ( $\sim 50k$  parameter) architecture. Under the null hypothesis  $H_0$  (Section 4), the dual-branch architecture neither helps nor hurts and the contribution reduces to the rigorous mainnet backtest of MCDM-EMA, the falsification of the LOB-to-DeFi transfer claim, and the open-source Extra+1/Extra+2 contributions of Section A — each of which stands on its own merits.

### 10.5 Three Layers of Cross-Domain Transfer

The empirical record now spans two structurally different markets: high-frequency LOB mid-price prediction [20] and the DeFi USDC lending-rate forecasting and allocation problem of Section 9. The cross-domain hypothesis behind both papers is not a single claim but a sequence of three: that an identical *architecture* carries signal across the two markets; that the resulting *predictions*



carry the same kind of out-of-sample structure; and that this in turn confers a *strategic* edge on an allocator built on top. The empirical record now lets us disaggregate the three claims, and the conclusion is that they fail at distinct layers — the architectural transfer succeeds, the predictive transfer is partial and regime-conditional, and the strategic transfer fails on the binding test window. We state each claim and the evidence in turn.

### 10.5.1 Architectural Transfer Succeeds

The DA-BiGRU-CNN of Section 4.1 trains stably and extracts non-trivial signal in both domains under *identical* network wiring, composite loss, and optimization schedule. In LOB mid-price prediction it attains test weighted-Pearson on the order of 0.74, outperforming a 205-feature LightGBM baseline by 0.5–1.6 percentage points [20]; in DeFi lending it reaches validation weighted-Pearson 0.747 on Aave with directional accuracy 0.739 and a positive Compound branch  $R^2 = 0.138$  (Section 9.2). The recipe that ports without modification is concrete and recordable: (i) identify a domain-natural decomposition pair  $(\mathcal{X}_A, \mathcal{X}_B)$  such that one subspace carries the deterministic backbone and the other carries the residual stochastic component — (price, volume) in LOB,  $(\varepsilon = r - f_{\text{kink}}(u), u + \text{ctx})$  in DeFi (Section 4.1); (ii) re-use the same dual-BiGRU stack with multi-scale Conv1d fusion (kernels  $\{3, 5, 7\}$ , hidden size 64); and (iii) re-use the same composite loss (Equation (2)) with weighted-Pearson as the dominant term. The *architectural* transfer hypothesis — “the same template generalises by re-identifying the decomposition pair” — is therefore supported by the validation diagnostics on both panels. This is the central methodological contribution of the present paper and is independent of the H1 outcome on the strategic layer.

A subsidiary but important architectural finding is that the kink prior has no LOB analogue and contributes a genuinely new lever in the DeFi setting. The governance-noise analysis of Section 10.1 shows that subtracting  $f_{\text{kink}}(u; \theta_{\text{current}})$  from the realized rate explains only 21.3% of supply-rate variance under static current parameters; conditioning on the protocol-published kink function absorbs a deterministic +186 bps offset that would otherwise be invisible to any architecture conditioning only on  $(r, u)$ . The LOB-to-DeFi transfer is in this sense *strictly enriching* at the architectural layer: every LOB-side architectural choice ports, and one DeFi-native lever (the kink prior) is added.

### 10.5.2 Predictive Transfer is Partial

The transfer breaks one layer up. Validation weighted-Pearson 0.747 on the Aave branch collapses to test  $-0.07$ , and directional accuracy on the cross-protocol label falls from 0.739 on validation to 0.351 on test (Section 9.2) — a sub-chance value whose sign is the empirical signature of a forecaster that has locked onto a correlation pattern from the wrong regime rather than of a forecaster that is uncertain. The proximate cause is documented and dated: between 2025-Q3 and 2025-Q4 the Aave-pays-more share rises from 39.9% to 56.3% and the median cross-protocol spread inverts from  $-0.218$  pp to  $+0.183$  pp (Section 10.2); the validation fold lives in the pre-inversion regime; the test fold straddles the inversion and a return to Compound dominance in 2026-Q1. The empirical contrast with the LOB benchmark of [20] is sharp. LOB microstructural dynamics on millisecond horizons are dominated by stationary adverse-selection mechanics whose statistical character does not shift on a four-month time scale; DeFi lending rates inherit on-the-order-of-quarter regime persistence from slow demand-side composition shifts (USDC borrow-demand cohort changes, perpetual DEX funding-rate cycles, periodic governance-set risk-parameter updates), and on the panel assembled here a single quarter is the characteristic regime time-scale rather than a sub-component of within-regime noise. Predictive transfer is therefore conditional on regime alignment, not architecturally guaranteed: in-distribution predictability does not imply robust short-horizon predictability in DeFi the way it appears to in microstructure.

A constructive reading of this partial-transfer result is the quarterly-conditional retraining schedule of Section 11.4. The same architecture, trained on a rolling pre-inversion window and re-fit at the Q3-to-Q4 boundary, is the obvious next experiment. The negative test-fold weighted-Pearson does not falsify the architecture; it falsifies the implicit assumption that a one-shot training pass on the pre-inversion regime gives a forecaster whose calibration survives the inversion.

### 10.5.3 Strategic Transfer Fails on This Window

The predictive MCDM under-performs the reactive MCDM-EMA by 8 bp in annualised net APY (3.36% vs 3.44%); the paired monthly-Sharpe bootstrap of Section 9.3 returns  $-900.40$  with 95% CI  $[-1799.57, -1.23]$  and  $p = 1.00$ , firmly inside the  $H_0$  acceptance region of the pre-registered  $\Delta\text{SHARPE} \geq 0.2$  test. The hysteresis-sweep ablation of Section 9.5 rules out the most natural rebuttal — “the threshold is mis-tuned” — by showing that the design  $\theta = 0.05$  sits at the Sharpe knee of the sweep, with fewer rebalances (and zero gas at  $\theta \in \{0.10, 0.20\}$ ) actually *not raising* net APY against EMA in this window. The mechanism is therefore: a forecaster whose test-fold directional sign is wrong cannot drive a switching allocator into the right pool ahead of the EMA-smoothed reactive comparator; the predictive overhead consumes gas without obtaining the timing alpha that justifies it.

This is the same negative-ensemble pattern that [20] §5.5 reported for the LOB setting — combining a strong dual-branch GRU forecaster with a 205-feature LightGBM ranker *degraded* test weighted-Pearson by 0.05 to 1.6 percentage points rather than improving it — generalised one layer up the stack: combining a forecaster with a reactive allocator can under-perform the allocator alone if the forecaster transports out-of-distribution badly. Two negative results, in two structurally different domains, with the same mechanistic explanation: *the apparent complementarity of two components is a property of the joint distribution they were calibrated on, not a robust architectural fact*. This is, in our reading, a non-trivial finding that the  $H_0$ -publishable protocol of Section 1.2 is designed to surface. A literature in which only  $H_1$ -positive ensembles are reported is a literature in which this cross-domain regularity is invisible.

### 10.5.4 Relation to Halpern–Pass–Saraf, Revisited

The three-layer disaggregation lets us refine the response to [8] previewed in Section 10.3. Their result is a no-arbitrage statement about equilibrium fairness at the contract-pricing layer; it constrains *which interest rate the protocol could in principle charge* under strategic borrowers. Our finding is at the execution layer and it is empirical rather than theoretical: short-horizon local predictability of the realized rate stream is present in-distribution (validation weighted-Pearson 0.747 on Aave) but does not survive the 2025-Q3-to-Q4 regime shift documented in the data panel. These two negative findings do not refute each other — one is a no-arbitrage statement about equilibrium fairness, the other is an out-of-sample statement about the stationarity of the short-horizon predictive structure — and they do not jointly imply that DeFi-lending forecasting is hopeless. They jointly imply that the empirical-predictability conjecture must be re-stated in a regime-conditional form, with explicit handling of distribution shift on the order of one fiscal quarter rather than a global predictability claim across an 18-month panel. The  $H_0$ -publishable verdict is in this sense complementary to [8]: it provides the empirical data point that the impossibility-of-fair-pricing result predicts will manifest as regime-persistent disequilibrium in the realised rate stream.

## 11 Conclusion and Future Work

### 11.1 Summary of Findings

We tested whether a domain-aware dual-branch recurrent network (DA-BiGRU-CNN) calibrated for LOB mid-price prediction [20] transfers — without architectural change — to DeFi USDC lending-rate forecasting on Aave V3 and Compound V3, and whether the resulting forecast yields a Sharpe improvement over a reactive EMA-smoothed MCDM allocator on a binding four-month held-out test window. Under the same composite loss (Equation (2)) and the substitution of the domain-natural decomposition pair from (price, volume) to  $(\varepsilon = r - f_{\text{kink}}(u), u + \text{ctx})$ , the network trains stably on the 13,096-hour Aave panel and attains validation weighted-Pearson 0.747 on Aave with directional accuracy 0.739 — a non-trivial in-distribution signal that confirms the architectural transfer hypothesis of Section 10.5.1. The pre-registered H1 test on the binding test fold, however, returns  $\Delta\text{SHARPE} = -900.40$  with 95% CI  $[-1799.57, -1.23]$  and  $p = 1.00$ : *fail to reject  $H_0$* . The hysteresis ablation (Section 9.5) rules out a mis-tuned threshold as the cause; the documented 2025-Q3-to-Q4 regime inversion of Section 10.2 is the proximate explanation.

### 11.2 Contribution at Three Layers

We report the negative H1 verdict honestly per the H0-publishable protocol pre-registered in Section 1.2; the contribution stands at three layers, each of which is independently defensible. At the *architectural layer*, identical wiring carries non-trivial signal across two structurally different financial markets (microsecond-resolution order-book microstructure and hour-resolution lending-rate dynamics), supporting the DA-BiGRU-CNN template as portable across the (price, volume) / (residual, utilization) decomposition pair. The kink-prior extension of Section 4.1 is, in our reading, the first architecturally-encoded use of a protocol-published rate model as a hard prior in a deep DeFi-rate forecaster. At the *infrastructure layer*, the open-source Compound V3 base-rate loader and the `BaseLendingAllocationStrategy` abstraction upstream-able to `fractal-defi` [12] (documented in Section A) are, to our knowledge, the first multi-protocol lending-allocation primitive in that substrate; both are ready for upstream PR independent of the H1 outcome. At the *empirical layer*, the falsified LOB-to-DeFi-lending strategic transfer — mirroring the “negative ensemble” finding of [20] §5.5 one layer up the stack (Section 10.5.3) — is itself publishable evidence that complements the equilibrium impossibility result of [8]: it provides the directly-measured short-horizon regime-shift instance that the impossibility result predicts will manifest in the realised rate stream (Section 10.5.4).

### 11.3 Limitations

Four limitations bound the present claims. (i) The 18-month panel covers less than two full macro cycles, and the binding test fold is four months, yielding only four paired monthly observations for the bootstrap of Section 9.3; a longer test fold would tighten the CI without changing the sign of the contrast. (ii) The two-protocol scope is deliberate but excludes Morpho, Spark, and cross-chain venues; cross-protocol structural diversity beyond Aave/Compound is untested. (iii) The single underlying (USDC) is intentional but limits external validity to the stablecoin family; cross-asset rate forecasting (ETH, wBTC, DAI) is plausibly more or less regime-stable. (iv) The forecaster is trained once on the pre-inversion window; online recalibration across regime boundaries (Section 10.5.2) is not implemented here.

### 11.4 Future Work

Four extensions are immediate and each directly addresses one limitation. First, *regime-conditional retraining*: re-fit the forecaster on a rolling pre-window at the Q3-to-Q4 boundary and re-run the H1 test; this is the cleanest empirical falsification of the regime-attribution argument

in Section 10.5.2. Second, *longer test folds*: extending the panel to a  $\sim 36$ -month window and re-running the four-month rolling H1 test gives a distribution of bootstrap outcomes against which the present single-window verdict can be properly contextualized. Third, *N-way protocol extension*: Morpho and Spark integration on the existing `BaseLendingAllocationStrategy` abstraction would test whether the predictive edge scales beyond the Aave–Compound pair. Fourth, *architectural ablation under regime-aware training*: an attention-backbone variant along the lines of [24] and [14] can be tested under the same regime-aware training schedule, isolating whether the present H1 negative is a DA-BiGRU-CNN capacity ceiling or a data-regime structural limit. A Panoptic-style options overlay on the supply position, hedging tail risk during the high-volatility tertile of Section 10.2, is under active exploration as a complement to (rather than substitute for) the forecasting layer.

## A fractal-defi PR Artifacts

**Extra+1: Compound V3 Lending Loader (with uniform utilization field).** A new `CompoundV3LendingLoader` pulls hourly `marketHourlySnapshots` from the Messari subgraph, normalizes WAD scaling to the same per-second continuous-compound convention as the upstream Aave V3 loader, and emits a `LendingHistory` record. As a uniformity fix, both Aave and Compound `GlobalState` objects gain a `utilization` field computed from `totalLiquidity` and `totalCurrentVariableDebt` in the existing subgraph response. PR title: “*Add Compound V3 lending history loader (Messari subgraph) and uniform utilization field on lending GlobalState*”.

**Extra+2: BaseLendingAllocationStrategy.** A new abstract strategy template in `fractal/strategies/` with four overridable hooks (`compute_criteria_vector`, `aggregate_criteria`, `select_target`, `should_rebalance`). The proposed `PredictiveMCDMStrategy` is the first concrete subclass. PR title: “*Add BaseLendingAllocationStrategy for multi-lending-protocol allocators*”.

## B Hyperparameter Grids

Full grids for both the forecaster (Section 6.1) and the strategy (Section 6.2) are logged in MLflow under `results/mlflow/`. The defaults reported in the body correspond to the configurations that maximize validation weighted-Pearson under the directional-accuracy floor and the turnover constraint, respectively.

## C Reproducibility

Code is available at <https://github.com/SergeySolovyev/predictive-mcdm-defi>. Python dependencies are pinned in `requirements.txt` (`fractal-defi==1.3.2`, see ERRATA in `README.md`). Environment variables `THE_GRAPH_API_KEY`, `DUNE_API_KEY`, and `FRAC TAL_DATA_PATH` configure the data fetchers. The full reproduction pipeline is the nine-step sequence in `README.md` §Quick start, ending in `latexmk -pdf main.tex`.

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